Depth-uniform Gaussian Bayesian learning of shallow Boolean targets

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Abstract

Consider fully connected tanh networks with an independent Gaussian prior, conditioned on absolute output margin greater than 1/2 on the Boolean cube and on noiseless training labels. Suppose the target has a one-hidden-layer representation of width C, weights bounded by R, and signed output margin at least one. For each fixed $\zeta > 0$, we prove a sample bound polynomial in input dimension, C, R, learner depth, inverse error, and logarithmic inverse confidence, simultaneously for every learner width above its depth-specific bounded-weight circuit complexity and every Gaussian exponent $\gamma \geq 1 + \zeta$. No internal robustness restriction or infinite-width limit is required. The result covers parity, all symmetric Boolean functions, and polynomial-size DNF/CNF. An extension covers targets of any fixed intrinsic depth, with polynomial degree depending on that intrinsic depth. More precisely, target depth D_0 costs a factor $(2 + d + C + R)^{O_\zeta(D_0)}$, with polynomial dependence on every learner depth $L \geq D_0$. The proof combines a Gaussian-distance decomposition, a width-independent tanh product-norm bound, and an energy-efficient representation obtained by zero-padding and scalar transmission. It does not establish the mean-field case $\gamma = 1$ or polynomial dependence on the size of an arbitrary deep representation.

1 Statement and conventions

The target $f : \{-1, 1\}^d \to \{-1, 1\}$ is learned from independent examples $(x_i, f(x_i))$, where x_i have an arbitrary fixed distribution D. A learner of depth $L \geq 2$ has $L - 1$ hidden layers, all of width N, and computes

$$h^0 = (1, x), \qquad h^s = \tanh(W^s h^{s-1}) \quad (1 \leq s < L), \qquad z_w(x) = v^T h^{L-1}.$$

There are no biases after the augmented input. Define

$$\kappa_N(w) = \frac{d}{N}\|W^1\|_F^2 + \sum_{s=2}^{L-1} \|W^s\|_F^2 + N\|v\|_2^2.$$

The raw Gaussian $\mu_{N,t}$ has independent entry variances $t^2/d, t^2/N, t^2/N^2$, respectively. In energy coordinates its density is proportional to $\exp[-N\kappa_N/(2t^2)]$. Write $\sigma^2 = t^2/N$. The exponent-γ convention is $t^2 = N^{1-\gamma}$, equivalently $\sigma^2 = N^{-\gamma}$.

The prior conditions this Gaussian on $|z_w(x)| > 1/2$ at every Boolean input. The posterior additionally conditions on $f(x_i)z_w(x_i) > 0$ at every training input. Thus it is equivalently the raw

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Gaussian conditioned on the intersection of these events. Throughout, "unrestricted" means that no additional internal robustness condition is imposed; it does not remove the original all-input output margin.

Let $C_L(f)$ be the least width N for which a depth-L representation has all weights bounded by N and signed output greater than $1/2$ on the entire cube. The main theorem, proved in Section 5, states that a width-C, weight-bound-R, unit-margin shallow target has sample complexity

$$\mathrm{poly}_\zeta(d, C, R, L, \varepsilon^{-1}, \log(1/\delta))$$

for every specified $L \geq 2$, simultaneously at all $N \geq C_L(f)$ and all $0 < \sigma^2 \leq N^{-(1+\zeta)}$. The sample event has probability at least $1 - \delta$, and posterior Gibbs error is at most ε. The polynomial degree may depend on ζ; no uniform limit as $\zeta \downarrow 0$ is asserted. No claim of computationally efficient posterior evaluation is made.

If $p_S(x)$ is posterior probability of the correct label, then

$$D\{x : p_S(x) \leq q\} \leq \frac{\mathrm{GibbsErr}(Q_S)}{1-q} \quad (0 < q < 1).$$

Hence the same theorem gives the probability-of-correct-answer thresholds $2/3$ and $1/2$ after multiplying the Gibbs error bound by three and two. Arbitrary mixtures of tied or untied posterior functions are allowed. For auxiliary classifiers, take $\mathrm{sign}(0) = +1$.

Proof organization and scope. A finite parameter grid handles polynomially bounded learner widths. At larger widths and $\gamma > 1$, a typical Gaussian weight array has small operator norms in normalized coordinates. Gaussian isoperimetry forces the posterior close to this typical set. Subtracting the typical part leaves a low-energy signal with exactly the same Boolean signs. Product-norm Rademacher control bounds its generalization error. Finally, a shallow representation can be padded and transmitted through any number of layers so that the relevant product-norm cost stays polynomial in its shallow parameters. All arguments and quantitative estimates needed for this result are included here. The draft has undergone independent AI-agent proof reviews but is not Lean-certified.

Related representation-cost questions. Jacot [1] studies the infinite-depth limit of normalized representation cost for homogeneous networks. Our scalar padding serves a related purpose, but tanh is nonhomogeneous and the required estimates also control finite Gaussian posterior mass at every admissible width. Depth separations for norm-controlled ReLU classes [2] concern another learning model; they do not by themselves settle the Gaussian-posterior question considered here.

2 Unrestricted learning at bounded widths

A robustness restriction is unnecessary when the learner's width is itself polynomially bounded. The following elementary argument retains the original all-input clean margin and Gaussian conditioning conventions, and is uniform in the Gaussian scale. It also covers analog target networks at their original depth, without digital simulation.

Let $P_N = N(d+2) + (L-2)N^2$ be the parameter count, and let $\mu_{N,t}$ denote the raw Gaussian with variances $t^2/d, t^2/N, t^2/N^2$, respectively. Thus $\kappa_N = (t^2/N)\chi^2_{P_N}$ under $\mu_{N,t}$. Fix integers $1 \leq C \leq M$ and suppose, for some $D \geq 1$, that

$$\mu_{N,t}\{w : f(x)z_w(x) > 1/2 \text{ for every } x\} \geq \exp(-DN/t^2). \tag{1}$$

This holds for $C \le N \le M$ and $0 < t \le 1$. Only this target-mass hypothesis is needed; posterior networks may be fragile.

Theorem 2.1. *For $0 < \varepsilon, \delta < 1$, set*

$$B = 4D + 2[d + 2 + (L-2)M]\log 2 + 4\log(2/\varepsilon),$$
$$A = \sqrt{MB}, \qquad T = d + 1 + M(L-1), \qquad q = [4T(MA)^{L-1}]^{-1}.$$

If

$$n \ge \frac{2}{\varepsilon}\left[\log M + P_M \log(5A/q) + \log(1/\delta)\right],$$

then, with probability at least $1-\delta$ over independent examples from any input distribution, the unrestricted Gaussian posterior has Gibbs error at most ε, simultaneously for every $C \le N \le M$ and $0 < t \le 1$. Here "unrestricted" retains the original conditioning on $|z_w(x)| > 1/2$ for all inputs, but imposes no internal robustness condition. The same sample event works after any number of additional correctly labeled examples. In particular, if D and M are polynomial in d, L, C, the sample bound is polynomial in $d, L, C, \varepsilon^{-1}, \log(1/\delta)$, uniformly for all $\gamma \ge 1$ with $t^2 = N^{1-\gamma}$.

Proof. For every sample, the event used to condition the raw Gaussian contains the target event in (1). The chi-square exponential moment therefore gives

$$Q_{S,N,t}\{\kappa_N > B\} \le \exp\left(\frac{DN}{t^2} + \frac{P_N}{2}\log 2 - \frac{NB}{4t^2}\right) \le \varepsilon/2.$$

The last inequality uses $P_N/N \le d + 2 + (L-2)M$, $N \ge 1$, and $t \le 1$. This bound holds for every sample, without a probabilistic qualification.

If $\kappa_N \le B$, every physical parameter has magnitude at most A. Round each to the nearest multiple of q, using at most $5A/q$ grid values per coordinate. For the maximum coordinatewise hidden-state discrepancies, boundedness and the Lipschitz property of tanh give

$$e_1 \le (d+1)q, \qquad e_s \le NAe_{s-1} + Nq, \qquad |z - z_{\text{grid}}| \le NAe_{L-1} + Nq.$$

Consequently, uniformly over Boolean inputs,

$$|z - z_{\text{grid}}| \le q[d + 1 + M(L-1)](MA)^{L-1} = 1/4.$$

Rounding thus preserves the entire Boolean function of every posterior network with energy at most B.

The union of these grid classifiers over $C \le N \le M$ has logarithmic cardinality at most $\log M + P_M \log(5A/q)$. A consistency union bound gives one sample event on which every consistent grid classifier has true error at most $\varepsilon/2$. Every good-energy posterior draw has the same function as such a classifier; the remaining posterior mass is at most $\varepsilon/2$. This proves the Gibbs bound. The finite class is independent of t, and consistency only becomes more restrictive as examples are added. □

The logarithm of the grid size satisfies

$$\log(5A/q) = \log(20AT) + (L-1)\log(MA).$$

The exponentially small parameter precision costs only a polynomial in L in this logarithm. The unresolved unrestricted problem is therefore uniformity at learner widths *larger than every polynomial in the specified target complexity and depth*; this theorem does not show those larger widths actually require more samples.

Corollary 2.2 (Same-depth bounded-weight analog targets)**.** *Suppose a width-C, depth-L tanh network realizes f with $f(x)z(x) \geq 1$ on every Boolean input and weights of magnitude at most $R \geq 1$. Fix $M \geq C$ and set*

$$\begin{aligned} K &= 2[d(d+1) + (L-1)C^2]R^2, \\ a &= [4(d+1+M(L-1))(MR)^{L-1}]^{-1}, \\ D &= \max\{1, K + [d+2+(L-2)M][2+\log(1/a)]\}. \end{aligned}$$

Then (1) holds with this D. Thus Theorem 2.1 supplies a polynomial sample bound at the original depth for every $C \leq N \leq M$, with dependence on R only through $\log R$ in the final sample bound. No robustness condition on the target implementation or prior support is needed. A target with the original strict margin $> 1/2$ satisfies the displayed hypothesis after doubling its readout.

Proof. Clone the C neurons into balanced nonempty groups of sizes n_i summing to N. Copy first-layer rows and replace internal and readout weights by W_{ij}/n_j and v_i/n_i. This preserves the exact function. Since $\max_i n_i / \min_i n_i \leq 2$, its energy is at most K, and its weights remain bounded by R. A physical-coordinate perturbation of magnitude at most a changes every output by at most $1/4$, by the recurrence above.

Use isotropic coordinates $y = (\sqrt{d}\, W^1, \sqrt{N}\, W^{\text{internal}}, Nv)$, so $y \sim \mathcal{N}(0, t^2 I)$ and $\|y_*\|^2 \leq NK$ for the cloned center. The standard-normal-coordinate box $|y_j/t - y_{*,j}/t| \leq a$ lies within the physical box, because all coordinate scalings are at least one and $t \leq 1$. Its probability is at least

$$\begin{aligned} &\left(\frac{2a}{\sqrt{2\pi}}\right)^{P_N} \exp\left[-\frac{1}{2}\sum_j (|y_{*,j}|/t + a)^2\right] \\ &\qquad \geq \exp[-NK/t^2 - P_N(2 + \log(1/a))] \geq \exp(-DN/t^2). \end{aligned}$$

Here $(u+a)^2 \leq 2u^2 + 2a^2$ and $a \leq 1$ suffice. Every network in this box has target margin at least $3/4$. □

This is an elementary finite-parameter learning result, rather than a new claim about the general VC dimension of neural networks. Its role is to isolate which width regime remains open under the exact prior conventions.

3 A width-independent tanh bound without robustness

We first establish the width-independent function-class bound used to control the low-energy signal extracted from a posterior draw. The proof accounts for the fact that tanh is not positively homogeneous.

The normalized-activation and exponential-moment method is closely related to Nakhleh–Nowak's treatment of deep neural variation spaces [5], which explicitly uses $\tanh(st)/s$ and slope discretization, and to the exponential peeling argument of Golowich–Rakhlin–Shamir [4]. We give a self-contained product-Frobenius version for the present Gaussian-energy conventions; we do not claim the underlying proof technique is new.

Theorem 3.1 (Product-Frobenius bound for tanh)**.** *Let $\mathcal{F}_{L,P}$ contain all scalar-output tanh networks with $L-1$ hidden layers, arbitrary finite widths, a linear readout, and product of layer Frobenius norms at most P. Biases are absent except through an augmented input. On every sample $x_1, \ldots, x_n$ with $\|x_i\| \leq R$,*

$$\widehat{\mathfrak{R}}_n(\mathcal{F}_{L,P}) \leq \frac{PR}{\sqrt{n}} \left[2 + \sqrt{2(L-1)\log\big(3 + n(L-1)^2\big)}\right].$$

The bound is independent of every hidden width and of the individual layer norms apart from their product.

Proof. If $R = 0$, every sampled input and every network output on the sample is zero, so the bound is immediate. Assume $R > 0$ below. Define $\phi_s(u) = \tanh(su)/s$ for $s > 0$, $\phi_0(u) = u$, and $\phi_\infty(u) = 0$. These functions are odd, 1-Lipschitz, and zero at zero. For finite $s, t \geq 0$ and $r \geq 0$,

$$\phi_s(ru) = r\phi_{sr}(u), \qquad |\phi_s(u) - \phi_t(u)| \leq u^2|s - t|, \qquad |\phi_s(u)| \leq 1/s \quad (s > 0).$$

The derivative bound follows by differentiating $\phi_s(u) = u \int_0^1 \operatorname{sech}^2(suv)\, dv$.

Let $\mathcal{G}_0$ contain the identity feature map. Inductively allow arbitrary finite-dimensional features

$$g_j(x) = \phi_{s_j}(a_j^T h(x)), \quad h \in \mathcal{G}_{\ell-1}, \quad \sum_j \|a_j\|^2 \leq 1, \quad s_j \in [0, \infty].$$

All these feature vectors have norm at most R on the sample. Put $B_\ell(\epsilon) = \sup_{g \in \mathcal{G}_\ell} \|\sum_i \epsilon_i g(x_i)\|$ for independent Rademacher signs. Writing $a_j = r_j u_j$ and using the scaling identity shows

$$B_\ell \leq \sup_{h,\, \|u\| \leq 1,\, s \in [0,\infty]} \left| \sum_i \epsilon_i \phi_s(u^T h(x_i)) \right|.$$

For any $\eta > 0$, uniformly on $[-R, R]$ the slope family admits an η/n approximation using at most $J = 3 + n^2R^2/\eta^2$ choices: truncate $s > n/\eta$ to the zero function, and round slopes below that threshold down at mesh $\eta/(nR^2)$. The error in the Rademacher sum is at most η.

We use the convex contraction principle

$$\mathbb{E}G\left(\sup_{t \in T} \sum_i \epsilon_i \psi_i(t_i)\right) \leq \mathbb{E}G\left(\sup_{t \in T} \sum_i \epsilon_i t_i\right)$$

for convex nondecreasing G and 1-Lipschitz ψ_i vanishing at zero. For completeness, the single-coordinate proof compares $X_\pm = \sup_t(a_t \pm \psi(t_k))$ and $Y_\pm = \sup_t(a_t \pm t_k)$. One has $\max(Y_+, Y_-) \geq \max(X_+, X_-)$ by $|\psi(t)| \leq |t|$. An approximately maximizing pair for X_+, X_- also gives $Y_+ + Y_- \geq X_+ + X_-$, by the Lipschitz property. Convex monotonicity therefore gives $G(Y_+) + G(Y_-) \geq G(X_+) + G(X_-)$; iterate over coordinates. Bounded separable classes follow by approximation from finite classes. In our application, u ranges symmetrically and ϕ_s is odd, so the absolute supremum is the same as the one-sided supremum.

Apply this principle with $G(t) = e^{\lambda t}$ to each grid slope, then bound the exponential of a maximum by the sum of the exponentials:

$$\mathbb{E}e^{\lambda B_\ell} \leq e^{\lambda\eta} J\, \mathbb{E}e^{\lambda B_{\ell-1}}.$$

Furthermore, $\mathbb{E}B_0 \leq R\sqrt{n}$, and bounded differences gives $\mathbb{E}e^{\lambda B_0} \leq e^{\lambda R\sqrt{n} + \lambda^2 nR^2/2}$. Writing $H = L - 1$, iteration and Jensen yield

$$\mathbb{E}B_H \leq H\eta + R\sqrt{n} + \frac{H \log J}{\lambda} + \frac{\lambda n R^2}{2}.$$

Take $\eta = R\sqrt{n}/H$ and optimize λ to obtain $\mathbb{E}B_H \leq R\sqrt{n}[2 + \sqrt{2H \log(3 + nH^2)}]$.

Finally, for a particular original network, write $M_\ell = \|W^\ell\|_F$ and $S_\ell = \prod_{j \leq \ell} M_j$. Dividing its ℓth hidden vector by S_ℓ changes its recursion to $\phi_{S_\ell}((W^\ell/M_\ell)g^{\ell-1})$. Thus its normalized hidden features belong to the enlarged classes above, and its scalar output is at most P times a norm-one linear readout of $\mathcal{G}_H$. Zero-norm layers give the zero function and require no normalization. This proves the theorem. All suprema are measurable by countable approximation of finite-dimensional parameter spaces and then union over widths. □

For the Gaussian energy in our model, the product of the L weighted squared layer norms is dP^2. Consequently

$$P \le d^{-1/2}(\kappa_N/L)^{L/2} \le d^{-1/2}e^{B/2} \qquad \text{if } \kappa_N \le L + B.$$

Applying the theorem to the ramp loss $\ell(yz) = \min\{1, \max\{0, 1 - 2yz\}\}$ gives the following explicit bound.

Corollary 3.2 (All widths at bounded excess energy)**.** *Fix $B \ge 0$. With probability at least $1 - \delta$, every sample-consistent depth-L network, at any finite width, with training margins at least $1/2$ and energy $\kappa_N \le L + B$ has sign error at most*

$$\frac{4\sqrt{2}e^{B/2}}{\sqrt{n}}\left[2 + \sqrt{2(L-1)\log(3 + n(L-1)^2)}\right] + \sqrt{\frac{\log(1/\delta)}{2n}}.$$

No internal robustness assumption is imposed.

Proof. Augmented Boolean inputs have norm $\sqrt{d+1}$, so $PR \le \sqrt{2}e^{B/2}$. The ramp loss is bounded by one and is 2-Lipschitz. Symmetrization, scalar contraction, and bounded differences give population loss at most empirical loss plus four times the network Rademacher bound and the stated confidence term. Empirical loss is zero and population loss upper-bounds sign error. □

Lemma 3.3 (A scalar transmission construction)**.** *For every real $D > 0$ and integer $m \ge 2$, there are positive scalar weights $\alpha, b_2, \ldots, b_{m-1}, c$ such that*

$$r_1 = \tanh\alpha, \quad r_s = \tanh(b_s r_{s-1}), \quad c r_{m-1} = 1/2,$$

and, writing $D' = \max\{D, 4\}$,

$$D\alpha^2 + \sum_{s=2}^{m-1} b_s^2 + c^2 \le (m+12)(D'/4)^{1/m}, \qquad c^2 = (D'/4)^{1/m}.$$

Proof. Set $q = (D'/4)^{1/(2m)}$, $k_s = \min\{s, m-s\}$, and $r_s = q^s/(\sqrt{D'}k_s)$. Then $0 < r_s \le 1/(2qk_s) \le 1/2$ and $r_{m-1} = 1/(2q)$. Choose $\alpha = \operatorname{atanh} r_1$, $b_s = \operatorname{atanh}(r_s)/r_{s-1}$ and $c = q$. For $0 < u \le 1/2$ the power series gives $(\operatorname{atanh}(u)/u)^2 \le 1 + u^2$. Writing $R_s = k_{s-1}/k_s$, the reciprocal pairs $k/(k+1), (k+1)/k$ satisfy

$$\left(\frac{k+1}{k}\right)^2 + \left(\frac{k}{k+1}\right)^2 - 2 \le 4/k^2.$$

Consequently $\sum_{s=2}^{m-1} R_s^2 \le m - 2 + 7$, $R_s \le 2$, and $\sum_{s=1}^{m-1} k_s^{-2} < 4$. Using $q^{2m} = D'/4$ therefore gives

$$D\alpha^2 \le q^2 + 1/4, \qquad \sum_{s=2}^{m-1} b_s^2 \le q^2(m - 2 + 7) + 4, \qquad c^2 = q^2.$$

Their sum is at most $(m+12)q^2$. Empty internal sums at $m = 2$ satisfy the same estimates. □

4 Unrestricted finite-variance learning for $\gamma > 1$

There is a stronger bridge than concentration of the full Frobenius energy. A typical Gaussian bulk has small operator norms when $\gamma > 1$, even when its total Frobenius energy is large. Gaussian isoperimetry shows that the posterior lies near such a bulk, with the remaining signal having energy close to a target witness's. Removing the bulk then preserves all Boolean signs under an explicit product-norm condition. No additional restriction of prior support is used.

Write $\sigma^2 = t^2/N$, so the original Gaussian has independent entry variances $N\sigma^2/d, \sigma^2, \sigma^2/N$. Its energy coordinates are

$$A_1 = \sqrt{d/N}\, W^1, \qquad A_s = W^s \ (2 \le s < L), \qquad A_L = \sqrt{N}\, v,$$

and all their entries are independent $\mathcal{N}(0, \sigma^2)$. Their squared Euclidean norm is exactly κ_N.

Theorem 4.1 (Unrestricted learning above exponent one)**.** *Fix integers $d \ge 1$, $L \ge 2$, $C \ge 1$, and a real $K \ge L$. Use tanh hidden activations and a linear readout. Suppose that for every $N \ge C$ there is a width-N, depth-L target witness satisfying*

$$f(x)z(x) \ge 1 \quad \textit{for every Boolean input}, \qquad \kappa_N \le K.$$

Put

$$Q_* = \left(\frac{K+2}{L}\right)^{L/2}.$$

For every fixed $\zeta > 0$, every $0 < \varepsilon, \delta < 1$, and every distribution D on the Boolean cube, the original Gaussian learner, conditioned only on the original global clean margin and noiseless training consistency, has a sample bound polynomial in

$$d, L, C, K, Q_*, \varepsilon^{-1}, \log(1/\delta)$$

which, with probability at least $1 - \delta$ over independent training inputs drawn from D and labeled by f, gives Gibbs error at most ε, simultaneously for all $N \ge C$ and all $0 < \sigma^2 \le N^{-(1+\zeta)}$. Thus it covers all $\gamma \ge 1 + \zeta$ at their actual finite Gaussian scales. The polynomial degree may depend on ζ; no uniform claim as $\zeta \downarrow 0$ is made.

We prove the statement with explicit width thresholds and sample inequalities. The hypothesis $K \ge L$ costs nothing: averaging over the cube and using AM–GM shows that a global unit-margin witness has $\kappa_N \ge Ld^{1/L} \ge L$.

A perturbation estimate using operator norms. Normalize hidden states by $\sqrt{N}$ and the augmented input by $\sqrt{d}$. The input norm is at most $\sqrt{2}$, all coordinate activations remain 1-Lipschitz and zero-preserving, and the layer maps are the A_s above. If U has energy at most $K + 1$, and an additive weight perturbation B has operator norm at most η in every normalized layer, telescoping one layer at a time gives

$$\sup_x |z_{U+B}(x) - z_U(x)| \le \sqrt{2}\, L\eta \left[\frac{K + 1 + 2\eta\sqrt{L(K+1)} + L\eta^2}{L-1}\right]^{(L-1)/2}. \tag{2}$$

Indeed, each summand is at most $\sqrt{2}\eta$ times a product of $L - 1$ other layer norms. The sum of squares of all perturbed layer-norm bounds is at most the numerator, and AM–GM bounds each such product. Set

$$A = \left(\frac{K+2}{L-1}\right)^{(L-1)/2}, \qquad \eta = \min\left\{1, \frac{1}{4\sqrt{L(K+1)}}, \frac{1}{8\sqrt{2}\, LA}\right\}.$$

The numerator in (2) is at most $K+2$, so the output change is at most $1/8$. Furthermore $A \le \sqrt{e}\, Q_*$, hence η^{-1} is polynomial in L, K, Q_*.

A symmetric typical Gaussian set. For completely explicit constants, put

$$a_d^2 = 2(d+1)\log 2 + 4\log 8, \qquad b_L^2 = 16\log 9 + 8\log(8L), \qquad b = \max\{a_d, b_L\}.$$

Let T consist of bulk matrices whose first and last energy-coordinate Frobenius norms are at most ta_d, and whose internal operator norms are at most tb_L. Then T is closed, convex and centrally symmetric, and

$$\mu_{N,t}(T) \ge 1/2. \tag{3}$$

For the first block, its squared Frobenius norm is $(t^2/N)\chi^2_{N(d+1)}$; the readout is the same expression with $d+1$ replaced by one. The exponential moment of χ^2 bounds each failure probability by $1/8$. For an internal matrix $tG/\sqrt{N}$, a $1/4$-net of the unit sphere has at most 9^N points, and the two-net scalar Gaussian bound gives

$$\Pr\{\|tG/\sqrt{N}\|_{\mathrm{op}} > tb_L\} \le 2\exp(2N\log 9 - Nb_L^2/8) \le \frac{1}{4L}.$$

Union bounding over the $L-2$ internal blocks proves (3). Every normalized block of a member of T has operator norm at most tb.

Target mass by a Gaussian shift. Suppose $tb \le \eta$. For a target witness w_*, the shifted set $w_* + T$ has target margin at least $7/8$, by (2). Central symmetry and the Gaussian change-of-mean identity yield

$$\begin{aligned}\mu(w_* + T) &= e^{-\kappa_N(w_*)/(2\sigma^2)} \int_T e^{-\langle w_*, B\rangle_\kappa/\sigma^2}\, d\mu(B) \\ &\ge e^{-K/(2\sigma^2)}\mu(T) \ge \tfrac{1}{2}e^{-K/(2\sigma^2)}.\end{aligned}$$

The inequality follows by pairing B with $-B$ and using $\cosh u \ge 1$. This lower bound holds for the raw probability of the conditioning event for every correctly labeled sample.

The posterior is close to the typical set. Let $d_\kappa(w, T)$ be Euclidean distance in energy coordinates. Gaussian isoperimetry [3] and (3) imply

$$\mu\{d_\kappa(w,T) > r\} \le e^{-r^2/(2\sigma^2)}.$$

Dividing by the preceding target-mass lower bound gives, for every sample,

$$Q_s\{d_\kappa(w,T)^2 > K+1\} \le 2e^{-1/(2\sigma^2)}. \tag{4}$$

On the complement, let B be the metric projection of w onto T and write $U = w - B$. The projection is well defined and continuous, since T is nonempty, closed and convex. Then $\kappa_N(U) \le K+1$ and $\sup_x |z_w(x) - z_U(x)| \le 1/8$. The original posterior has clean margin $> 1/2$ everywhere, so U has *exactly the same Boolean function* as w and training margins $> 3/8$. It is enough to use a margin-$1/4$ ramp loss below.

A polynomial width split. Take $\beta = \varepsilon/2$ and

$$M = \left\lceil \max\left\{C, 2, (b/\eta)^{2/\zeta}, 2\log(2/\beta)\right\}\right\rceil.$$

For every $N \geq M$ and $\sigma^2 \leq N^{-(1+\zeta)}$, one has $t \leq N^{-\zeta/2}$, hence $tb \leq \eta$, and $2e^{-1/(2\sigma^2)} \leq \beta$. The product norm of U is at most $d^{-1/2}[(K+1)/L]^{L/2}$. Write

$$G = \sqrt{(d+1)/d}\left(\frac{K+1}{L}\right)^{L/2}, \qquad H = L - 1.$$

The product-norm Rademacher bound, now with the 4-Lipschitz ramp loss $\min\{1, \max\{0, 1-4yz\}\}$, gives one sample event of probability at least $1-\delta/2$ on which every such consistent residual network has error at most

$$F_n = \frac{8G}{\sqrt{n}}\left[2 + \sqrt{2H\log(3+nH^2)}\right] + \sqrt{\frac{\log(2/\delta)}{2n}}. \tag{5}$$

This event already includes all finite widths. When $F_n \leq \varepsilon/2$, adding the posterior exceptional mass β proves the required bound for every $N \geq M$ and every stated scale.

For $C \leq N \leq M$, use Theorem 2.1. Here is a target-mass constant requiring only the present witness assumption. Every witness weight is at most $R = \sqrt{MK}$. Put

$$a = [4(d+1+M(L-1))(MR)^{L-1}]^{-1},$$
$$D_M = \max\{1, K + [d+2+(L-2)M][2+\log(1/a)]\}.$$

The same Gaussian-coordinate box calculation as in the finite-width proof has physical perturbation at most a, retains target margin at least $3/4$, and has mass at least $\exp(-ND_M/t^2)$. Applying that theorem with error ε and failure probability $\delta/2$ controls all these widths and all $t \leq 1$ simultaneously. Its finite-grid logarithm is polynomial in $M, d, L, \log K$. The union of the two sample events proves the theorem. Finally, $G \leq \sqrt{2}Q_*$, $b^2 = O(d + \log L)$ and $A \leq \sqrt{e}Q_*$ show that M and a sample size satisfying all the displayed inequalities are polynomial in the stated arguments for fixed ζ. This proves the asserted quantitative dependence.

Scope. This argument does not apply at $\gamma = 1$: the Gaussian bulk's operator norm does not tend to zero with width there. It also does not make Q_* polynomial for every small circuit. If a fixed base witness can be padded with energy $K = L + B$, where B is independent of the added depth, then $Q_* \leq e^{(B+2)/2}$. The theorem therefore removes exponential dependence on *additional depth*, at every width and every fixed exponent above one, while retaining a potentially exponential dependence on base-witness energy. Stronger target-specific energy estimates can make Q_* itself polynomial.

5 Polynomial unrestricted learning of shallow targets at every depth

A shallow target can have a small first-layer Gaussian energy at a polynomial base width simply by adding zero neurons. Its readout energy is then polynomial. Distributing that readout cost over the learner's remaining layers makes the product-norm parameter polynomial uniformly in depth. This gives a family for which neither arbitrary width nor arbitrary added depth requires a robustness restriction.

Theorem 5.1 (Shallow targets, all depths and all admissible widths)**.** *Suppose a Boolean target $f : \{-1,1\}^d \to \{-1,1\}$ has a one-hidden-layer representation*

$$z(x) = \sum_{j=1}^{C} v_j \tanh(a_j^T \bar{x}), \qquad \bar{x} = (1, x), \qquad f(x)z(x) \geq 1$$

with every coordinate of a_j and every v_j bounded in absolute value by $R \geq 1$. Let $C_L(f)$ denote its bounded-weight network complexity at depth L: the least width N admitting a realization with all weights at most N in absolute value and signed output greater than $1/2$ everywhere.

For each fixed $\zeta > 0$, the original unrestricted Gaussian learner has sample complexity polynomial in

$$d, C, R, L, \varepsilon^{-1}, \log(1/\delta),$$

for every specified learner depth $L \geq 2$ and every input distribution, simultaneously for every width $N \geq C_L(f)$ and every $\gamma \geq 1 + \zeta$. With probability at least $1 - \delta$ over a sample of that size, posterior Gibbs error is at most ε simultaneously at all these widths and all prior scales $0 < \sigma^2 \leq N^{-(1+\zeta)}$. The polynomial's degree may depend on ζ. The prior is conditioned only on the original global clean margin and the training labels.

Proof. Let A have rows a_j^T, and set

$$M = \left\lceil \max\{4, C, 2R, d\|A\|_F^2\} \right\rceil, \qquad T = \max\{1, M\|v\|_2^2\}.$$

Both are polynomial in d, C, R. Add zero hidden neurons to obtain width M. This preserves the function, and its first-layer and readout energies satisfy

$$E_{\text{pre}} = \frac{d}{M}\|A\|_F^2 \leq 1, \qquad c^2 = M\|v\|_2^2 \leq T.$$

In fact $M\|v\|_2^2 \geq 1$, since unit margin and $|\tanh| \leq 1$ imply $\|v\|_2 \geq 1/\sqrt{C}$, but the maximum makes this unneeded.

For any $N \geq M$, balanced cloning of this shallow network preserves its output exactly and gives

$$E_{\text{pre}} \leq 2, \qquad c^2 = N\|v_N\|_2^2 \leq 2T.$$

Explicitly, assign each of the M original neurons either $\lfloor N/M \rfloor$ or $\lceil N/M \rceil$ clones, copy its incoming weights, and divide its readout weight by its number of clones. The first cost increases by at most $M\lceil N/M \rceil / N \leq 2$, and the readout cost by at most $N/(M\lfloor N/M \rfloor) \leq 2$. For learner depth $L = 2$ we already have a unit-margin witness with

$$K_2 = 2 + 2T, \qquad Q_*^2 = \left(\frac{K_2 + 2}{2}\right)^2 = (T+2)^2.$$

For $L \geq 3$, apply Lemma 3.3 with $D = 4c^2$ and $m = L - 1$. Replace the old readout by a layer whose every row is $2\alpha v_N^T$. Follow it with the scalar transmission layers, implemented by equal coordinates with dense matrix entries b_s/N, and final readout entries q/N, where

$$q^2 = \max\{c^2, 1\}^{1/(L-1)} \leq (2T)^{1/(L-1)}.$$

The target had margin at least one, hence in particular at least $1/2$, so the scalar construction gives margin at least $1/2$. Double its final readout to restore margin at least one. This adds $3q^2$ to the energy. Thus every width $N \geq M$ has a depth-L unit-margin witness with

$$\kappa_N \leq K_L := 2 + (L + 14)(2T)^{1/(L-1)}.$$

Writing $q_*^2 = (2T)^{1/(L-1)} \geq 1$, we have $K_L + 2 \leq (L+18)q_*^2$. Consequently

$$Q_*^2 = \left(\frac{K_L+2}{L}\right)^L \leq e^{18}(2T)^{L/(L-1)} \leq e^{18}(2T)^{3/2} \qquad (L \geq 3). \tag{6}$$

The exponent $L/(L-1)$ is at most $3/2$ for $L \geq 3$. Theorem 4.1, with base width M, therefore provides a sample bound polynomial in the stated quantities for all $N \geq M$.

We verify admissible smaller widths rather than silently identifying M with the target's minimum complexity. First, $C_L(f) \leq M$ for all $L \geq 2$. At width M, the original incoming weights have magnitude at most $R \leq M/2$. In the added scalar channel, $\alpha \leq \operatorname{atanh}(1/2) < 0.55$, so its first weights $2\alpha v_j$ have magnitude at most $1.1R < M$. The scalar construction has $b_s \leq 2.2q$ and final weights $2q/M$. Since $T \leq MCR^2 \leq M^4$ and $L \geq 3$,

$$q \leq (2T)^{1/4} \leq 2^{1/4}M, \qquad b_s/M < 3 \leq M, \qquad 2q/M < 3 \leq M.$$

Thus this depth-L, width-M circuit satisfies the bounded-weight convention and has margin at least one. The case $L = 2$ is immediate.

For $C_L(f) \leq N < M$, start from a defining width-$C_L(f)$ circuit, clone it to width N, and double its readout. It has margin greater than one, all weights at most $2M$, and Gaussian energy bounded by a polynomial in d, L, M. The Gaussian box estimate and Theorem 2.1 consequently give a sample bound polynomial in $d, L, M, \varepsilon^{-1}, \log(1/\delta)$, simultaneously for all these widths and all $t \leq 1$. Use failure probability $\delta/2$ for this event and $\delta/2$ for the preceding event for $N \geq M$. Their intersection proves the claim. □

The complexity parameter controlling this theorem's sample size is the size and weight bound of a *shallow* representation. Its width admissibility is the actual depth-L complexity $C_L(f)$. The theorem does not identify shallow complexity with the minimum complexity of an arbitrary deep circuit, and therefore does not settle the unrestricted problem for all small deep circuits. It also makes no claim at $\gamma = 1$.

Corollary 5.2 (Parity at arbitrary depth and admissible width)**.** *For every fixed $\zeta > 0$, parity of any $k \geq 1$ chosen coordinates in $\{-1,1\}^d$ has unrestricted Gaussian posterior sample complexity polynomial in $d, k, L, \varepsilon^{-1}, \log(1/\delta)$, for every $L \geq 2$, every $N \geq C_L(f)$, and every $\gamma \geq 1 + \zeta$. The bound holds for every input distribution and for the original finite-variance posterior.*

Proof. Let $q = (k - \sum_{i \in S} x_i)/2 \in \{0, \ldots, k\}$ count negative coordinates, and put $a = \log(16k)$. Define

$$\tau_j(x) = \tanh(a(q - j + 1/2)), \qquad 1 \leq j \leq k.$$

Every argument has magnitude at least $a/2$, so $|\tau_j - \operatorname{sign}(q - j + 1/2)| \leq 2e^{-a} = 1/(8k)$. Writing $c_k = 1$ for even k and $c_k = 0$ for odd k, the exact step identity is

$$(-1)^q = c_k + \sum_{j=1}^{k} (-1)^j \operatorname{sign}(q - j + 1/2).$$

Hence $F = c_k + \sum_j (-1)^j \tau_j$ satisfies $|F - (-1)^q| \leq 1/8$. Represent the constant by one extra hidden neuron with augmented-input weight $\operatorname{atanh}(1/2)$ and no other incoming weights. A linear readout equal to $2F$ has signed margin at least $7/4$. Its threshold-neuron readout weights are $2(-1)^j$, and its constant-neuron weight is $4c_k$. Its incoming weights have magnitude at most $a(k+1)$. Thus $C = k + 1$ and $R = \max\{4, a(k+1)\}$ are polynomial in k. Apply Theorem 5.1. □

Corollary 5.3 (Symmetric Boolean targets)**.** *The conclusion of Corollary 5.2 holds for any Boolean function of the Hamming weight of k selected coordinates.*

Proof. If its values are $h(0), \dots, h(k) \in \{-1, 1\}$, then

$$h(q) = \frac{h(0) + h(k)}{2} + \sum_{j=1}^{k} \frac{h(j) - h(j-1)}{2} \operatorname{sign}(q - j + 1/2).$$

Each coefficient has magnitude at most one. Replacing the signs by the same τ_j and doubling the output gives margin at least $7/4$, using the same width and weight bounds as in the parity proof. □

Corollary 5.4 (Polynomial-size DNF and CNF targets)**.** *A Boolean function on d bits represented by a DNF or CNF with $s \geq 1$ clauses has unrestricted Gaussian posterior sample complexity polynomial in $d, s, L, \varepsilon^{-1}, \log(1/\delta)$, for every specified $L \geq 2$, every $N \geq C_L(f)$, and every $\gamma \geq 1 + \zeta$, with $\zeta > 0$ fixed.*

Proof. For a DNF, let $q_j(x)$ be the number of false literals in clause j, and put $a = \log(16s)$ and $\tau_j(x) = \tanh(a(1/2 - q_j(x)))$. The exact clause indicator is $[1 + \operatorname{sign}(1/2 - q_j)]/2$. Thus

$$H(x) = s - 1 + \sum_{j=1}^{s} \operatorname{sign}(1/2 - q_j(x))$$

is -1 when no clause is satisfied and at least 1 otherwise. Replacing the signs by τ_j changes H by at most $1/8$. Doubling the resulting expression gives global signed margin at least $7/4$. It is a shallow tanh network with $s + 1$ features: the extra feature is the constant $1/2$, with readout weight $4(s - 1)$. After removing duplicate literals and discarding contradictory clauses, each clause has at most d literals. Each q_j is affine in the bits, so the remaining incoming and readout weights have magnitude at most $R = \max\{4s, a(d + 1)\}$. The shallow-target theorem applies. A CNF is the negation of a DNF in negated literals, and negating the readout preserves every bound. Constant functions are immediate. □

5.1 Targets of any fixed intrinsic depth

The shallow-target argument extends to any fixed target depth by using different numbers of clones at its successive hidden layers. The degree of the resulting polynomial may depend on the target depth, but not on the learner's additional depth.

Corollary 5.5 (Fixed target depth, arbitrary learner depth)**.** *Fix an integer $D_0 \geq 2$ and $\zeta > 0$. Suppose f has a width-C, depth-D_0 tanh realization in the present architecture, with signed output at least one on the Boolean cube and all weights bounded by $R \geq 1$. For every specified learner depth $L \geq D_0$, the original unrestricted Gaussian learner has sample complexity polynomial in*

$$d, C, R, L, \varepsilon^{-1}, \log(1/\delta).$$

The degree and constants may depend on D_0 and ζ. For every input distribution, one sample event of probability at least $1 - \delta$ gives Gibbs error at most ε simultaneously for every $N \geq C_L(f)$ and every $0 < \sigma^2 \leq N^{-(1+\zeta)}$. Here $C_L(f)$ is the actual bounded-weight network complexity defined in Theorem 5.1. The prior retains the original global clean-margin conditioning and sign-only training likelihood.

Proof. Put $H = D_0 - 1$. Pad the target's hidden layers to width C with zero neurons if necessary, and write $W^1, \dots, W^H, v$ for its weights. Choose an integer

$$B = \left\lceil \max\left\{4, 2R, \frac{d}{C}\|W^1\|_F^2, \|W^2\|_F^2, \dots, \|W^H\|_F^2\right\}\right\rceil, \qquad M = CB^H,$$

omitting the internal-layer terms when $H = 1$. Since $\|W^1\|_F^2 \le C(d+1)R^2$ and $\|W^s\|_F^2 \le C^2R^2$ internally, B is polynomial in d, C, R, and so is M for fixed D_0.

At hidden layer s, give each original neuron exactly

$$n_s = B^{H-s}$$

active copies, and fill the other positions up to width M with zero neurons. Copy first-layer rows. For $s \ge 2$, give every edge from a copy of source j to a copy of target i weight W^s_{ij}/n_{s-1}. All unused edges are zero. Since $n_H = 1$, retain the original readout on the final active neurons. This preserves the function exactly. The first-layer and internal energy terms are respectively

$$\frac{d}{M} n_1 \|W^1\|_F^2 = \frac{d}{CB}\|W^1\|_F^2 \le 1, \qquad \frac{n_s}{n_{s-1}}\|W^s\|_F^2 = \frac{1}{B}\|W^s\|_F^2 \le 1.$$

Thus the energy before the readout is at most H. Put

$$T = \max\{1, M\|v\|_2^2\} \le MCR^2.$$

It is polynomial in d, C, R for fixed D_0.

Balanced cloning of this width-M network into any $N \ge M$ preserves the function and gives

$$E_{\text{pre}} \le 2H, \qquad c^2 = N\|v_N\|_2^2 \le 2T.$$

If $L = D_0$, these are already unit-margin witnesses with $K = 2H + 2T$, and

$$Q_*^2 = \left(\frac{K+2}{D_0}\right)^{D_0} \le (4T)^{D_0}.$$

If $L > D_0$, set $m = L - H \ge 2$. Replace the old readout by the scalar transmission from Lemma 3.3, using $D = 4c^2$, and double the final scalar readout to obtain unit margin. Its readout cost before doubling is $q^2 = \max\{c^2, 1\}^{1/m} \le (2T)^{1/m}$. The total energy is at most

$$K_L = 2H + (m+15)(2T)^{1/m}.$$

This is the desired depth: the H old hidden layers and $m-1$ new hidden layers total $L-1$. Since $(2T)^{1/m} \ge 1$,

$$\begin{aligned} Q_*^2 &= \left(\frac{K_L+2}{L}\right)^L \\ &\le \left(1 + \frac{H+17}{L}\right)^L (2T)^{L/(L-H)} \\ &\le e^{H+17}(2T)^{(D_0+1)/2}. \end{aligned}$$

The last step uses $L/(L-H) = 1 + H/m \le 1 + H/2$. For $L \ge 2D_0$, the sharper upper bound $Q_*^2 \le e^{H+17}(2T)^2$ also holds. Both K_L and Q_* are polynomial in the stated parameters when D_0 is fixed, uniformly over the learner depth. Theorem 4.1 now controls every $N \ge M$.

It remains to check admissible widths below M. First $C_L(f) \le M$ at every $L \ge D_0$. The tapered construction has all original weights at most $R \le M/2$. In a padded scalar channel, $\alpha \le \operatorname{atanh}(1/2) < 0.55$, so its first added weights $2\alpha v_j$ have magnitude less than M. The scalar construction has $b_s < 2.2q$, and

$$T \le MCR^2 \le M^4, \qquad q \le (2T)^{1/4} \le 2^{1/4}M.$$

Its dense internal entries b_s/M and final readout entries $2q/M$ are therefore less than $3 \le M$. This gives a bounded-weight width-M witness with unit margin at each learner depth; for $L = D_0$ no scalar channel is needed.

For $C_L(f) \le N < M$, clone a defining width-$C_L(f)$ realization to width N and double its readout. Its margin is greater than one and all weights are at most $2M$. The Gaussian box estimate and Theorem 2.1 give a bound polynomial in $d, L, M, \varepsilon^{-1}, \log(1/\delta)$ for all these widths and all $t \le 1$. Allocate failure probability $\delta/2$ to this event and to the preceding event for $N \ge M$. Their intersection proves the claim. □

Thus polynomial dependence on learner depth holds for targets of any fixed intrinsic depth with polynomial target-size dependence. The width $M = CB^{D_0-1}$ and the exponents above explain why this does not give a single polynomial jointly in the intrinsic target depth and target size. The mean-field exponent $\gamma = 1$ is also still outside this result.

Corollary 5.6 (Quantitative dependence on intrinsic depth)**.** *Fix $\zeta > 0$. There are constants $a_\zeta, b_\zeta > 0$, independent of the intrinsic depth, with the following property. Suppose f has a width-C, depth-D_0 tanh realization with unit signed margin and weights bounded by $R \ge 1$, where $D_0 \ge 2$, and set $S = 2 + d + C + R$. For every specified learner depth $L \ge D_0$, the sample bound*

$$n \ge \left\lceil S^{a_\zeta D_0} \left(L + \varepsilon^{-1} + \log(2/\delta)\right)^{b_\zeta} \right\rceil$$

suffices for posterior Gibbs error at most ε, with probability at least $1-\delta$ for every input distribution, simultaneously for all $N \ge C_L(f)$ and $0 < \sigma^2 \le N^{-(1+\zeta)}$. The prior and likelihood are those of Corollary 5.5.

Thus the dependence on intrinsic depth is $\exp[O_\zeta(D_0 \log S)]$, while the dependence on learner depth remains polynomial with degree independent of D_0. In particular, if $D_0 \le c \log S$ for a fixed constant c, the target-parameter factor is at most $\exp[O_{\zeta,c}((\log S)^2)]$. This is a quasipolynomial bound in S, not a single polynomial jointly in intrinsic depth and target size.

Proof. First suppose $L \ge 2D_0$. Use the tapered construction in the preceding proof, with $H = D_0 - 1$, $M = CB^H$, and $T = \max\{1, M\|v\|_2^2\}$. Its explicit definition gives $B \le S^5$, so $M \le S^{5D_0}$ and $T \le MCR^2 \le S^{7D_0}$. Put $m = L - H$. Since $L \ge 2D_0$, one has $L/m \le 2$ and

$$K_L = 2H + (m+15)(2T)^{1/m} = O(LT), \qquad Q_*^2 \le e^{H+17}(2T)^{L/m} \le e^{H+17}(2T)^2 = S^{O(D_0)}.$$

For the remaining range $D_0 \le L < 2D_0$, do not taper the target. Instead pad its original realization to $M_0 = \lceil \max\{4, C, 2R\} \rceil \le S^2$, and put $T_0 = \max\{1, M_0\|v\|_2^2\} \le S^5$. Balanced cloning to every $N \ge M_0$ gives

$$E_{\mathrm{pre}} \le 2HS^4, \qquad c^2 \le 2T_0.$$

At $L = D_0$, retain the readout. For $L > D_0$, append the same scalar transmission with $m = L - H \ge 2$. These constructions give, respectively,

$$K_L \le 2HS^4 + 2T_0, \qquad K_L \le 2HS^4 + (m+15)(2T_0)^{1/m}.$$

In either case $K_L \le 19LS^5$. Therefore

$$Q_*^2 \le (20S^5)^L \le S^{7L} \le S^{14D_0}.$$

The width-M_0 witness has weights bounded by M_0: the original weights satisfy $R \le M_0/2$, and the scalar-channel check from the preceding proof applies because $T_0 \le M_0^4$. Thus $C_L(f) \le M_0$ in this range.

In both depth ranges, the polynomial in Theorem 4.1 has degree and constants depending only on ζ. Substituting the corresponding bounds gives the stated sample form above the respective base width M or M_0. For smaller admissible widths, the defining-witness and Gaussian-box argument in the preceding proof gives a sample bound polynomial in $d, L, M, \varepsilon^{-1}, \log(1/\delta)$ (or with M_0 in place of M), with degree independent of D_0. Substitution gives the same sample form. Split the failure probability between the two width ranges and enlarge a_ζ, b_ζ to finish. □

6 Limitations

The result separates learner depth from shallow target complexity. A target that has only an exponentially large shallow representation need not receive a polynomial bound in the size of a much smaller deep representation. The unrestricted general small-deep-circuit problem therefore remains open. The mean-field exponent $\gamma = 1$ also remains outside the Gaussian-distance argument: its typical hidden Gaussian bulk has order-one operator norm. This failure of one proof is not a sample lower bound. The present result retains the original all-input output-margin conditioning; removing that condition would require another argument or a change of prediction and training conventions.

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